

Dynamic Health Insurance Claim Processor

Technical Specification, Dynamic Policy Verification & Realistic ML Evaluation Report

1. Executive Summary & Policy Standing Intelligence

The **Dynamic Health Insurance Claim Processor** provides real-time AI adjudication for medical claims. It features live **Dynamic Policy Standing Verification** (checking ACTIVE, INACTIVE, SUSPENDED, and FRAUD_FLAGGED policy statuses directly from database & dataset registries) before granting instant auto-approvals (<10ms) or escalating high-risk claims to an investigator workbench.

2. Realistic Machine Learning Model Evaluation Metrics

Evaluated on 200 evaluation claim records from the **Kaggle / CMS Medicare Claims Dataset** with natural noise & overlap:

- **Realistic Claims Noise Introduced:** Simulates overlapping boundary conditions, clinical variance, and unobserved factors.
- **Test Set Performance:** 83.00% Accuracy, 94.59% High Fraud Recall.

Evaluation Metric	Score Achieved	Benchmark Evaluation & Interpretation
Accuracy	83.00%	166 / 200 noisy test claims correctly classified.
Precision	84.34%	High precision ensuring low false accusations.
Recall	94.59%	Captures 94.59% of fraudulent/abusive claims.
F1-Score	0.8917	Strong harmonic balance between Precision and Recall.
ROC-AUC	0.8299	Robust risk probability discrimination capability.

3. FastAPI REST Gateway Integration

HTTP Method	Endpoint	Integration Purpose & Action
GET	/api/v1/health	Returns API service status & engine metadata.
GET	/api/v1/policies/verify/{policy_id}	Queries DB & dataset policy standing (ACTIVE, SUSPENDED, FRAUD_FLAGGED).
POST	/api/v1/claims/evaluate	Evaluates claim JSON payload & returns dynamic route decision.
GET	/api/v1/claims/samples	Pre-loads test claim presets (Active, Blacklisted, Suspended).
GET	/api/v1/metrics	Exposes live model metrics (83.00% Accuracy, 0.8299 ROC-AUC).

4. Dynamic Decision Routing Matrix

Target Route Queue	Adjudication Criteria & Thresholds	Operational Action
■ AUTO_PROCESSED	Composite Risk < 0.35 AND Model Confidence >= 0.20 AND Policy Active AND Zero Critical Flags.	■ Emergency Instant Approval (<10ms).
■ PENDING_ADDITIONAL_VALIDATION	0.35 <= Composite Risk <= 0.65 OR Model Confidence < 0.20.	Enqueued for secondary validation.

■ HUMAN_INVESTIGATION	Composite Risk > 0.65 OR Critical Flag (Blacklisted Policy, Sanctioned Provider, Duplicate Claim).	■ High-Priority Fraud Audit Workbench.
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